

PALLOB BARMON

Computer Science Graduate — IT, Software & Data

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SUMMARY

I am a Computer Science and Engineering graduate with a practical foundation in programming, databases, software development, data analysis, and modern IT technologies. I enjoy understanding how systems work, solving technical problems, and learning new tools when a project requires them. During my degree, I worked on software and data-driven projects and led the technical implementation of an A+ undergraduate thesis involving data processing, machine learning, deep learning, model development, and experimentation. I also have experience with Python, SQL, web technologies, Git/GitHub, and database systems. I am looking for an entry-level opportunity where I can contribute to a team, continue developing my technical skills, and grow into a strong IT or software professional.

TECHNICAL SKILLS

Programming: Python, SQL (proficient) · Java, C, JavaScript, React (familiar)

Machine Learning: Random Forest, SVM, XGBoost, Extra Trees — GridSearchCV tuning, class-imbalance handling, feature importance analysis

Deep Learning: U-Net, U-Net++, DeepLabV3+ (PyTorch, ResNet-50 encoder) — semantic segmentation, custom loss functions, GPU training

GIS & Remote Sensing: Google Earth Engine, Sentinel-1 SAR, Sentinel-2, FABDEM/GLO-30 DEM, Rasterio, GeoPandas, OpenCV, Shapely, OSMnx

Data & Backend: NumPy, Pandas, Matplotlib, scikit-learn, MySQL, SQLite, REST APIs, Git/GitHub

Cloud: Core cloud computing concepts; self-directed AWS learning roadmap in progress

FINAL YEAR THESIS — TOPOGRAPHIC OBJECT DETECTION, DHAKA

Comparative Analysis of Random Forest and U-Net for Topographic Object Detection Using Satellite Imagery

Eastern University | Co-authored (2-person team) — led technical implementation: data pipeline, feature engineering, model training & experimentation | Graded A+

- Engineered a **23-band multi-modal feature stack** (Sentinel-1 SAR, Sentinel-2, spectral indices, GLCM texture, temporal NDVI, elevation) covering 233.7 km² of Dhaka using Google Earth Engine.
- Showed data engineering outweighed algorithm choice: **22.4-point accuracy gain** from label cleaning and class consolidation vs. ~0-point gain from switching Random Forest → U-Net.
- Identified a counter-intuitive negative result — OBIA post-processing **cut accuracy by 6 points** — because Dhaka's median building footprint is roughly one satellite pixel, disproving a standard literature assumption for South Asian cities.
- Ran a 4-zone urban morphology analysis, finding U-Net's advantage over Random Forest **scaled with planning regularity** (+7.1pp in planned Uttara vs. +1.6pp in organic Old Dhaka).
- Independently extended the thesis into a **7-model benchmark** (RF, SVM, XGBoost, Extra Trees, U-Net, U-Net++, DeepLabV3+), identifying U-Net++ as the top performer at 93.4% accuracy.

CERTIFICATION

Front-End Development (React / Node.js / Vue.js / Angular)

IIT, Jahangirnagar University — EDGE Project, Bangladesh Computer Council · 80 hrs · Issued 29 Jun 2025

EDUCATION

B.Sc. in Computer Science & Engineering — Eastern University

2022 – 2026 (Graduated June 2026) · CGPA: 3.32 / 4.00

LINKS

LinkedIn: <https://www.linkedin.com/in/pallab-barman02/> | GitHub: <https://github.com/Pallabbarmon>